INCH-POUND

MIL-D-14978A (AR)
AMENDMENT 5
30 July 1998
SUPERSEDING
AMENDMENT 4
03 May 1989

MILITARY SPECIFICATION

DETONATOR, STAB, M55
LOADING, ASSEMBLING AND PACKING

MIL-D-14978 WAS INACTIVATED ON 10 July 1996 FOR NEW DESIGN

This amendment forms a part of MIL-D-14978A (AR), dated 30 October 1972, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1: SPECIFICATIONS, MILITARY: Add the following:

"MIL-L-10547 – Liner, case, and sheet, Overwrap, Water Vaporproof or Waterproof, Flexible"

PAGE 2

* 2.1: Under STANDARDS, MILIATRY, Delete "MIL-STD-1235 – Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes"

Paragraph 2.1

Delete: "ET - #8798331 – Equipment Tabulation"

2.1:
DRAWINGS: Add the following:

"8794342 - Seal, Metallic, Assembly, Details and Instructions for Application
8796522 - Marking Diagram and Sealing for Wood Packing Boxes
8799112 - Marking diagram and Sealing for Interplant or CONUS Shipment of Ammunition Fiberboard Boxes
9258182 - Box, Set-up, Packing, Ammunition, Inner, for Small Explosive Components
9258183 - Box, Set-up, Alternative, Packing, Ammunition, Outer, for Small Explosive Components

AMSC N/A **FSC 1375**

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9258184 - Box, Alternative, Packing, Ammunition, for Small Explosive Components"

PUBLICATIONS: Add the following new reference:
"Navy MK95 PD WS 4997"

PAGE 3

3.4.1 Delete this paragraph in its entirety and substitute:

"3.4.1 Sensitivity and energy output. The detonators shall function satisfactorily as evidence by a depression in the steel disc of a minimum depth of 0.010 inch when tested as specified in 4.4.3.2."

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4.2.2.1 Delete this paragraph in its entirety and substitute:

"4.2.2.1 Sensitivity and energy output (see 3.4.1). Three hundred fifteen (315) detonator assemblies shall be subjected to this test. If four (4) or more assemblies fail to comply with the applicable requirement, the first article sample shall be rejected. The test shall be performed as specified in 4.4.3.2."

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* 4.3.2 Delete third sentence "Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD1235 may be used if approved by the Government." and substitute the following:

"Sampling plans for major and minor defects shall be in accordance with MIL-STD-105."

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4.3.2.2 Major defects. Delete "103" in its entirety.

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4.3.2.3 Critical defects. Delete "1. and 4." in their entirety and substitute the following:

"1. Felt cushion missing (not applicable to alternate cork cushion)."

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4. Slide cover assembly reversed (not applicable to alternate design)."

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4.3.3.3.1 Delete this paragraph in its entirety and substitute the following:

"4.3.3.3.1 Sensitivity and energy output. Major defect (see 3.4.1) Sampling shall be in accordance with MIL-STD-105, Normal Inspection, Table III-A, using a sampling code letter P and an AQL of 0.40%. The provisions of paragraph 8 of MIL-STD-105, regarding reduced or tightened inspection shall apply. Testing shall be as specified in 4.4.3.2 (Code No. 09001)."

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Paragraph 4.3.4;

Delete: "ET - #8798331 – Equipment Tabulation"

PAGE 12

Add the following new paragraphs:

4.4.1.1 Determination of moisture content of lead azide, alternate method (see 4.3.3.1). The procedure is the same as paragraph 4.4.1 with the following exception. The sample shall be dried in a vacuum oven at a temperature of 95 + 5°C and a pressure of 80 ± 10mm of mercury for 45 minutes.

4.4.2.1.1 Moisture content, alternate method (see 4.3.3.2). The procedure is the same as paragraph 4.4.2.1 with the following exception. The sample shall be dried in a vacuum oven at a temperature of 60 ± 5°C and a pressure of 80 ± 10mm of mercury for 45 minutes.

4.4.3.1 Delete this paragraph in its entirety.

4.4.3.3 Delete this paragraph in its entirety and substitute the following:

"4.4.3.3 Waterproofness. The detonators shall be immersed in water to a depth of not less than two inches but not more than three inches for a period of 48 hours, minimum. Both the water and the detonators shall be within the temperature range of 60° to 80°F throughout the test. Within one hour after removal from the water, the detonators shall be tested according to the test procedure for functioning (see 4.4.3.2)."

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Delete Section 5 in its entirety and substitute the following:

"5. PACKAGING

5.1 Preservation and packaging.
PAGE 16 (cont)

5.1.1 Level A.

5.1.1.1 Unit package. The detonators shall be preserved and packaged in accordance with dwg 9258182, sheets 1 and 2.

5.1.1.2 Intermediate packaging. The detonators shall be preserved and packaged in accordance with dwg 9258183, sheets 1 and 2.

5.1.2 Level B – same as level A.

5.1.2.1 Unit package. Same as level A.

5.1.2.2 Intermediate package. Same as level A.

5.1.3 Level C – same as a level A.

5.1.3.1 Unit package. Same as level A.

5.1.3.2 Intermediate package. Same as level A.

5.2 Packing.

5.2.1 Level A. The detonators shall be packed in accordance with dwg 9258184, sheets 1 and 2 except Box, Packing shall be Grade A, Type II, Class 1. A metallic seal (dwg 8794342) in both boxes is mandatory. A caseliner conforming to Type I, Grade A material, Specification MIL-L-10547 is required when cushioning is used between the exterior box and outer set-up boxes. Appropriate size fillers for End, Side, Bottom and Top shall be placed between the caseliner and exterior box. After inserting required number of out set-up boxes in liner, close and seal in accordance with Specification MIL-L-10547.

5.2.2 Level B. Same as level A.

5.2.2.1 Unit pack. Same as level A.

5.2.2.2 Intermediate pack. Same as level A.

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5.2.3 Level C.

5.2.3.1 Unit package. Unit package shall be in accordance with dwg 9258182.

5.2.3.1.2 Intermediate package. Intermediate package shall be in accordance with dwg 9258184, sheets 1 and 2.

5.2.3.1.3 Alternative. Alternatives No. 1 and No. 2. Exterior Packing shall be in accordance with dwg 9258184, sheets 1 and 2.

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5.3 Marking.

5.3.1 Level A. Level A shall be marked, stamped and sealed in accordance with dwg 8796522 and instructions shown on dwg 9258184, sheets 1 and 2.

5.3.2 Level B. Same as level A.

5.3.3 Level C. Box and Alternative Box No. 1 shall be marked and strapped in accordance with dwg 8796522. Box, alternate No. 2 shall be marked and sealed in accordance with dwg 8799112.

5.4 Shipping. When components from more than one lot are shipped as a carload, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of lots in transit."

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Add the following new paragraphs:

"6.6 Navy procurement orders. Manufactured to Army spec MIL-D-14978A, additionally incorporating environmental test required per paragraph 4.2.3, 4.2.4 and figure 3 of the MK95 purchase description WS 4997 with skip lot acceptance modified as follows: Upon successful testing of the initial lot, begin periodic environmental skip lot testing using a group of five lots, and subsequently using groups of nine lots – should a lot fail, reliability must be reestablished by successfully testing two successive lots prior to resumption of skip lot testing as described above.

6.6.1 Detonator identification. Those "Navy M55 detonators manufactured in accordance with paragraph 6.6 will be

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specifically identified prior to shipping – method to be determined by manufacturing AAP.

6.6.2 Test report distribution. The “Navy” M55 detonator test reports including lot acceptance tests shall be distributed to: Naval Surface Weapons Center, White Oak, Silver Spring, MD (NSWGWQ) codes G43 and G44, 20907 and to Pacific Missile Test Center, Pt Mugu, CA code 2141, 93042.

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